Supervised Learning (Part 1)

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Supervised Learning

A type of learning where the model is trained on labeled data.

Examples:

- Classification (e.g. Logistic Regression, Support Vector Machines)
- Regression (e.g. Linear Regression)

Understanding Data

A dataset is a collection of data that is organized in a structured format.

Key Components of a Dataset:

1. Instances (Rows) → Individual data points or observations.
2. Features (Columns) → Attributes or variables describing each instance.
3. Labels (Optional) → The target variable in supervised learning.

Understanding Data

Age	Height (cm)	Weight (kg)	Fit/Unfit
25	170	68	1
30	165	75	0
22	180	72	1
40	160	85	0
28	175	70	1

Understanding Data

- **Target/Label**: The column that we want to predict. It is our resultant column and we want to know for future data. In this dataset, it's the fit/unfit column marked in green.

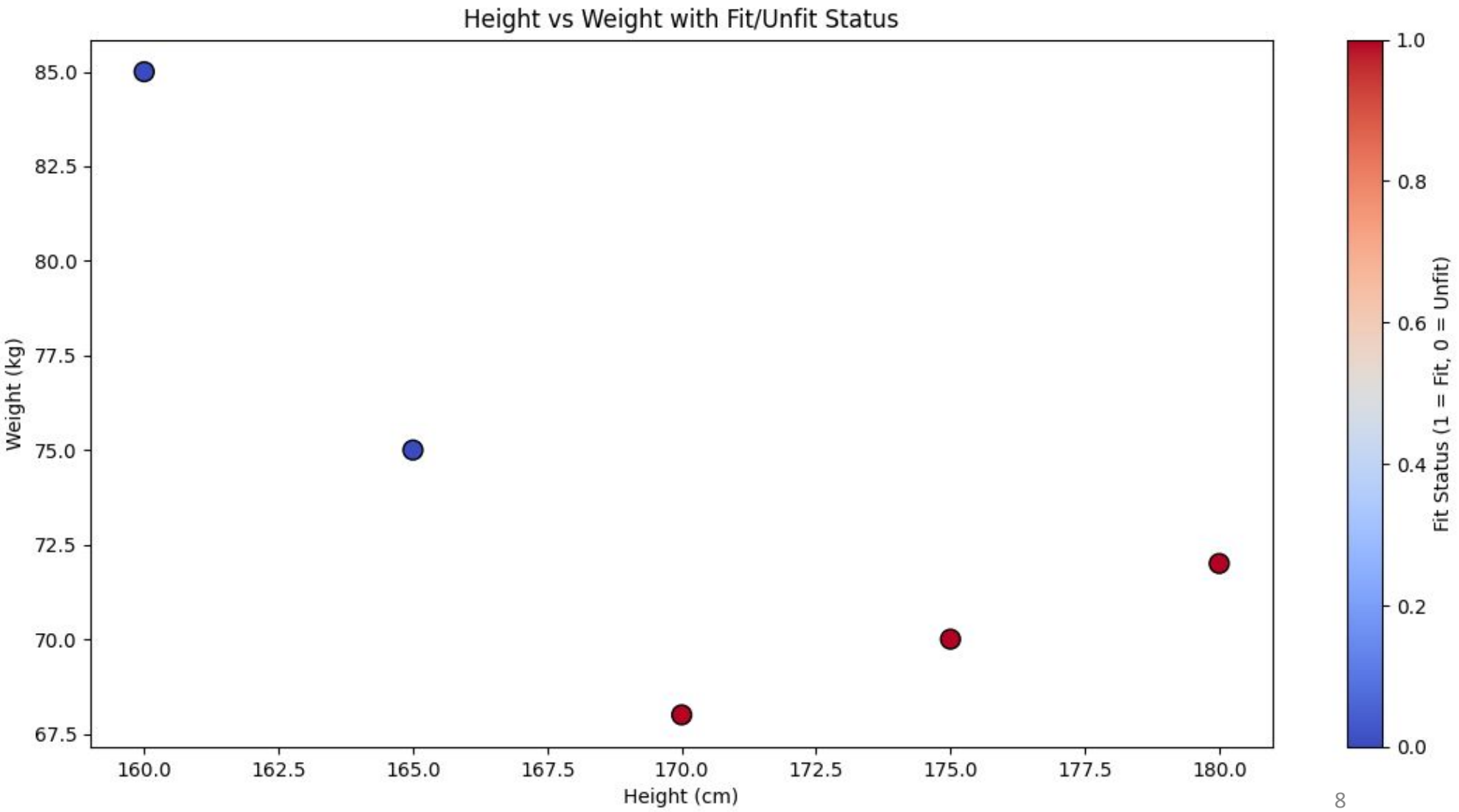
- **Variables/Features:** The columns other than the Target column. These columns help the ML model to predict the target for future data points. In this dataset, the variables are age, height, and weight.

k Nearest Neighbors (kNN)

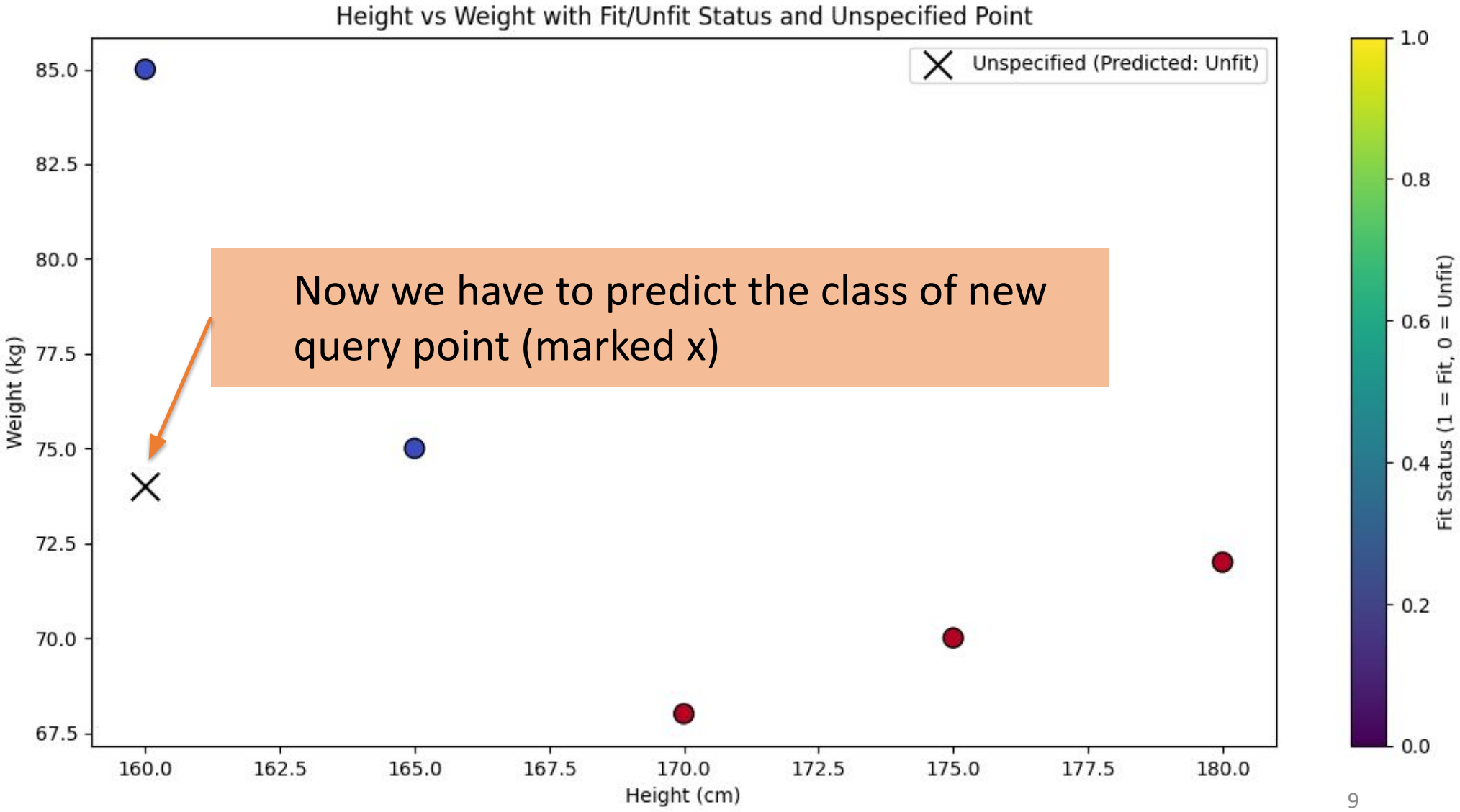
k-Nearest Neighbor (k-NN)

- Used to solve both classification and regression problems.
- Also known as an **instance-based model** or a **lazy learner** because it doesn't construct an internal model.
- Finds the **k nearest neighbors** and predict the class by the **majority vote** of the nearest neighbors.

k-Nearest Neighbor (k-NN)



k-Nearest Neighbor (k-NN)

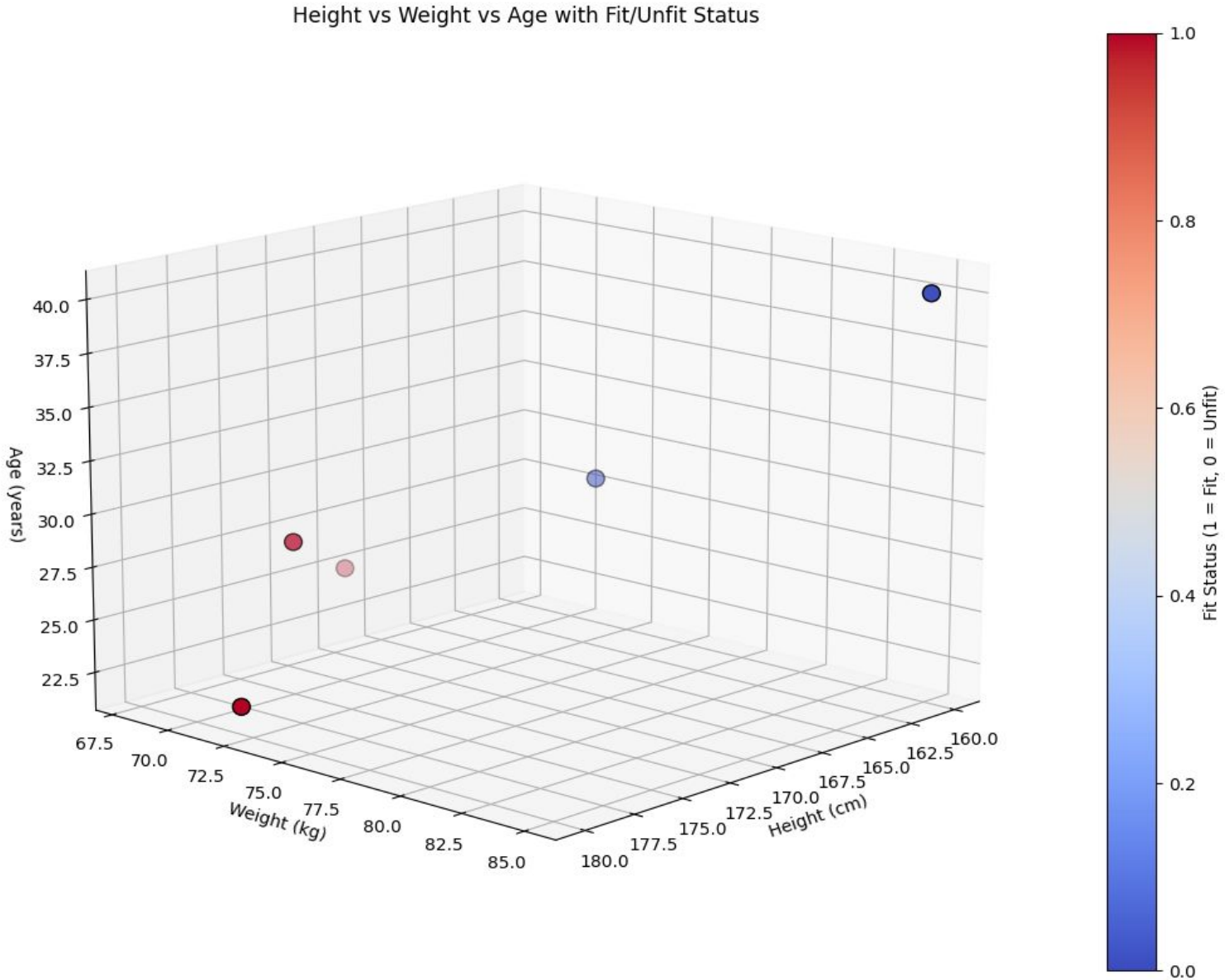


Now we have to predict the class of new query point (marked x)

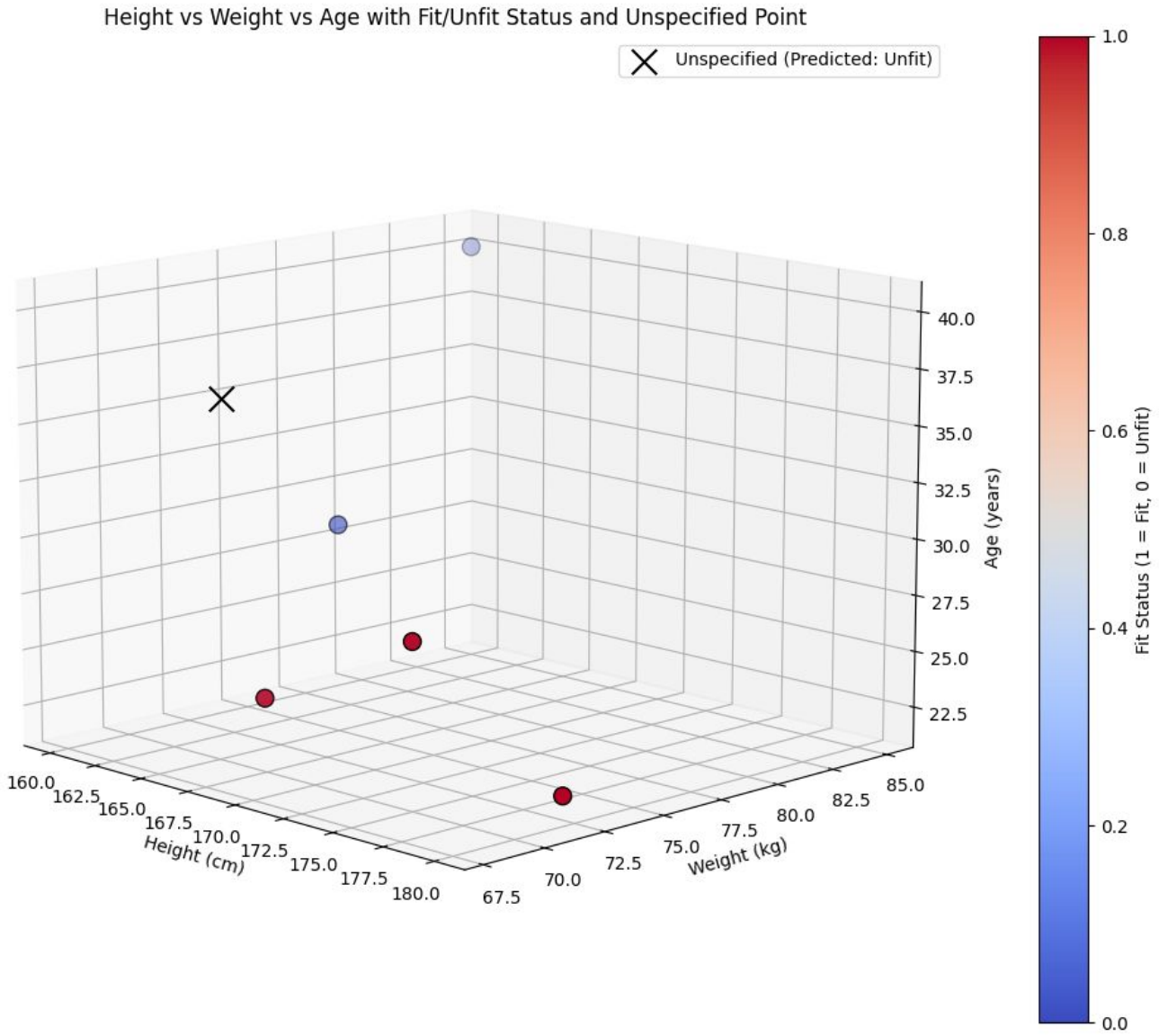
k-Nearest Neighbor (k-NN)

- Suppose, the value of k is 3
- 2 of the nearest points are unfit
- So, the query point belongs to the class unfit

k-Nearest Neighbor (k-NN)

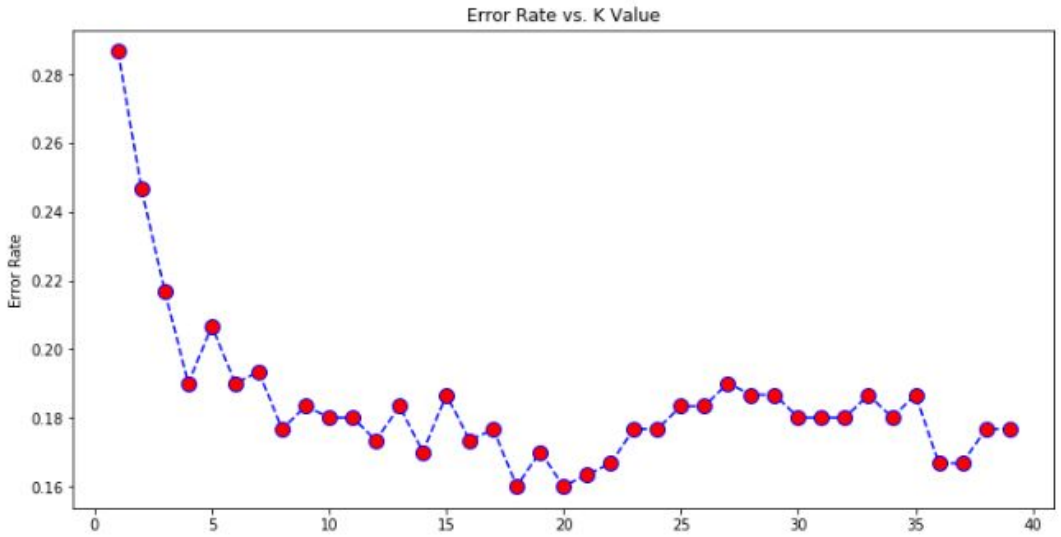


k-Nearest Neighbor (k-NN)



Finding K

- If we choose k =1 means the algorithm will be sensitive to outliers.
- If we choose k= all (means the total number of data points in the training set), the majority class in the training set will always win.
- Always use k as an odd number.
- Elbow Method: We can also use an error plot or accuracy plot to find the most favorable K value.



Finding K nearest neighbors

- The nearest neighbors are determined based on the distance of available points from the query point
- Types of distance:
  - Euclidean
  - Manhattan
  - Minkowski

Euclidean Distance

- For two points A(x_1, y_1) and B(x_2, y_2) Euclidean distance is -
- $d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Manhattan Distance

- Also known as City Block distance
- For two points $A(x_1, y_1)$ and $B(x_2, y_2)$ distance is -
- $d(A, B) = |x_2 - x_1| + |y_2 - y_1|$

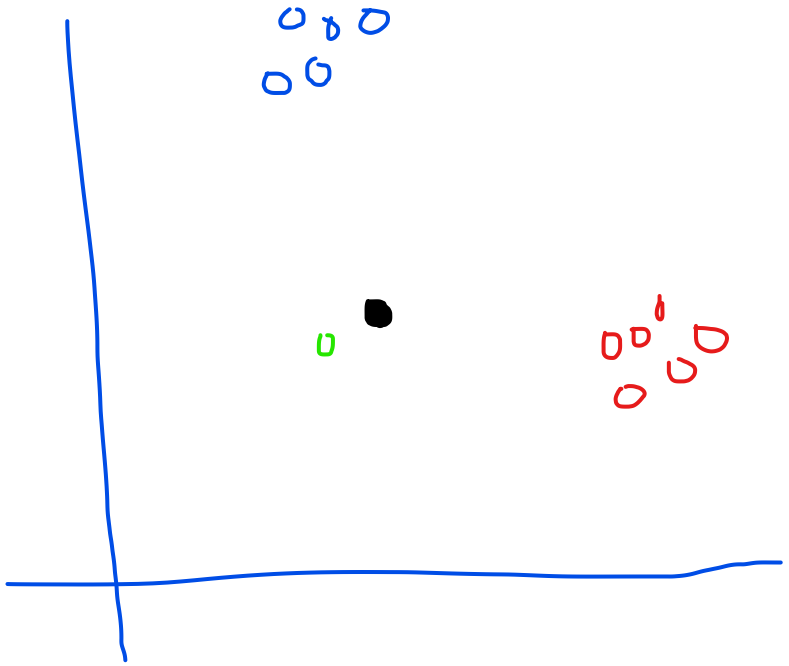
Minkowski Distance

- It is a generalized form of both Euclidean and Manhattan distances.
- For two points A(x_1, y_1) and B(x_2, y_2) distance is -
- $d(A,B) = \left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{\frac{1}{p}}$

- If p = 2, it's Euclidean Distance.
- If p = 1, it's Manhattan Distance.

Pros and Cons of KNN

- Pros
  - No training period
  - Easy implementation
  - Adding new data doesn't affect the model
- Cons
  - Does not work well for large datasets
  - Slower for high dimensions
  - Sensitive to noise
  - Feature scaling is mandatory



Practice Problem

🔹 **Query Point: (Age = 32, Height = 169 cm, Weight = 76 kg)**

📌 Goal: Predict if the person is **Fit (1) or Unfit (0)** using **k-NN**.

Age	Height (cm)	Weight (kg)	Fit/Unfit
25	170	68	1
30	165	75	0
22	180	72	1
40	160	85	0
28	175	70	1
35	168	78	0
27	182	74	1
38	158	90	0
29	172	71	1
33	166	80	0

Questions?